

Task 3 – Model Validation, Overfitting Control & Hyperparameter Tuning

AI / ML Internship | California House Price Prediction

Objective

Task 3 builds on Task 2 by adding a layer of validation that is usually skipped in beginner projects. Rather than just comparing test-set scores, this task looks at whether models are actually learning generalizable patterns or just memorizing training data. The three main additions are overfitting analysis, 5-fold cross-validation, and systematic hyperparameter tuning via GridSearchCV.

Overfitting Analysis

Overfitting happens when a model performs well on training data but poorly on new data. The gap between train RMSE and test RMSE is the clearest way to spot it. The table below shows the baseline results before any tuning:

Model	Train RMSE	Test RMSE	Gap	Test R ²
Linear Regression	66,959	67,385	+426	0.6535
Ridge (alpha=1)	66,959	67,385	+426	0.6535
Decision Tree (def.)	0	72,949	+72,949	0.5939
Random Forest (def.)	18,356	49,643	+31,286	0.8119

The Decision Tree is the most obvious example. A train RMSE of 0 means it memorized every training sample perfectly, but its test RMSE of 72,949 was actually worse than Linear Regression. That is textbook overfitting. Random Forest had a gap too, but its test performance was already the best of the group, which is why it was the main focus for tuning.

Cross-Validation Results

A single train-test split can be misleading depending on which samples land in the test set. 5-fold CV uses every sample for both training and validation, giving a more stable estimate. Each model was evaluated on the full scaled dataset with KFold(n_splits=5, shuffle=True, random_state=42).

Model	CV RMSE Scores	Mean RMSE	Std
Linear Regression	67385 / 66724 / 65409 / 64941 / 73325	67,557	3,014
Ridge (alpha=1)	67385 / 66724 / 65411 / 64940 / 73319	67,556	3,012
Decision Tree (def.)	72949 / 68733 / 70719 / 68327 / 69414	70,028	1,670
Random Forest (def.)	49662 / 48662 / 49494 / 46769 / 49865	48,890	1,136

Random Forest had both the lowest mean CV RMSE and the lowest standard deviation across folds, making it the most consistent performer. Linear Regression and Ridge were very similar to each other, as expected since the dataset does not strongly violate linearity assumptions. Decision Tree confirmed its tendency to overfit even before tuning.

Hyperparameter Tuning

GridSearchCV was used to tune three models. All tuning was done with 5-fold CV scoring so that the selected hyperparameters are chosen based on generalisation performance, not training accuracy.

Model	Parameters Searched	Best Params	Best CV RMSE
Decision Tree	max_depth, min_samples_split, min_samples_leaf	depth=10, split=20, leaf=5	58,998
Ridge	alpha: 0.01 to 500	alpha=50.0	67,534
Random Forest	n_estimators, max_depth, max_features	n=200, depth=None, feat=sqrt	49,288

For the Decision Tree, constraining depth to 10 and requiring at least 20 samples to split a node significantly reduced the train-test gap while improving test performance. Ridge barely changed from the default, which makes sense since linear models are already relatively stable on this dataset. Random Forest benefited most from using 200 estimators and sqrt feature sampling.

Final Model Comparison

Results on the held-out test set after tuning, compared against Task 2 baselines:

Model	MAE	RMSE	R ² Score
Random Forest (Task-3 tuned)	\$32,901	\$49,302	0.8145
Decision Tree (Task-3 tuned)	\$40,721	\$61,265	0.7136
Linear Regression (Task-3)	\$48,243	\$67,385	0.6535
Ridge Regression (Task-3 tuned)	\$48,221	\$67,414	0.6532
Random Forest (Task-2 baseline)	\$33,916	\$51,562	0.7971
Linear Regression (Task-2 baseline)	\$48,243	\$67,385	0.6535
Ridge Regression (Task-2 baseline)	\$48,242	\$67,386	0.6534

Final Model Selection and Justification

The tuned Random Forest was selected as the final model for the following reasons:

Best test performance: Lowest RMSE of \$49,302 and highest R² of 0.8145 on the held-out test set.

Most consistent CV scores: Standard deviation of 1,136 across folds, which was the lowest of all models tested. A low std means the score is not the result of a lucky split.

Overfitting controlled: Using max_features=sqrt means each tree only sees a random subset of features, which reduces correlation between trees and keeps the ensemble from overfitting. The 200-estimator ensemble also averages out noise that would trip up a single tree.

Improvement over Task 2: R^2 improved from 0.797 in Task 2 to 0.814 after tuning, showing that proper hyperparameter selection does make a measurable difference even for an already decent model.

CV-based selection: The best hyperparameters were chosen by GridSearchCV based on 5-fold CV RMSE, not training accuracy, which is the correct way to tune.

Ridge and Linear Regression performed almost identically, which suggests the dataset does not have a severe multicollinearity or overfitting problem for linear models. The Decision Tree improved with tuning but still could not match the Random Forest because individual trees are inherently higher variance than an ensemble of them.

Model saved as task3_best_house_price_model.pkl for future use.